

2024 AMC 10B Problem 16 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 16

**Problem:**

Jerry likes to play with numbers. One day, he wrote all the integers from 1 to 2024 on the whiteboard. Then he repeatedly chose four numbers on the whiteboard, erased them, and replaced them by either their sum or their product. (For example, Jerry's first step might have been to erase 1, 2, 3, and 5, and then write either 11, their sum, or 30, their product, on the whiteboard.) After repeatedly performing this operation, Jerry noticed that all the remaining numbers on the whiteboard were odd. What is the maximum possible number of integers on the whiteboard at that time?

Answer Choices:

- A. 1010
- B. 1011
- C. 1012
- D. 1013
- E. 1014

Solution:

Each time this operation was performed, the number of even integers on the whiteboard was reduced by at most 3. There were $\frac{2024}{2} = 1012$ even integers on the whiteboard initially. Because $\frac{1011}{3} = 337$ and $\frac{1014}{3} = 338$, Jerry needed at least 338 operations to eliminate all of them. After 338 operations there were $2024 - (338 \cdot 3) = (A) \boxed{1010}$ numbers on the whiteboard.

To see how all the even numbers could have been eliminated using these operations, suppose Jerry chose $6n + 1$, $6n + 2$, $6n + 4$, and $6n + 6$ and replaced them by their sum, for $0 \leq n \leq 336$. For the final operation, Jerry could have erased 3, 5, 9, and 2024 and replaced them with their sum. Then after these 338 operations, all 1010 numbers on the whiteboard were odd.

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